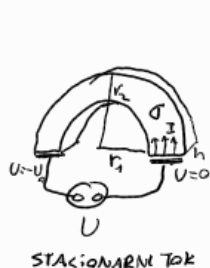


1. Upornost prevodne ploščice



r_1, r_2, h, σ

$R = ?$

$$R = \frac{U_0}{I}$$

RA:

1) $U(r, \varphi) = 0$

2) $U(r, \pi) = -U_0$

3) $j_{\perp} = j_r|_r = 0$
4) $j_r|_r = 0$

$$\vec{j} = \sigma \vec{E} = -\sigma \nabla U$$

$$\nabla \cdot \vec{j} + \frac{d\rho}{dt} = 0$$

$$\nabla \cdot \vec{j} = 0 \rightarrow \sigma \cdot (-\nabla^2) U = 0 \rightarrow \boxed{\nabla^2 U = 0}$$

splošna rešitev:

$$U(r, \varphi) = \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) (C_m r^m + D_m r^{-m}) + \underbrace{(a\varphi + b)}_{\text{člen } m=0} (c \ln r + d)$$

1) $\Rightarrow C_m = D_m = 0, c = 0$

$\Rightarrow U(r, \varphi) = A\varphi + B$ $I = \int j_{\varphi} dS = \sigma \frac{U_0}{h} \int \frac{h dr}{r} = \frac{\sigma U_0 h}{\pi} \ln \frac{r_2}{r_1}$

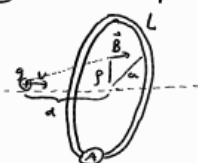
1) $B = 0$

2) $A = -\frac{U_0}{\pi}$

$$j_{\varphi} = -\sigma \frac{1}{r} \frac{\partial U}{\partial \varphi} = \sigma \frac{U_0}{\pi} \frac{1}{r}$$

$$R = \frac{U_0}{I} = \frac{\pi}{\sigma h \ln \frac{r_2}{r_1}}$$

2. Cabreraov eksperiment



(magnetni monopol)

v, a, L, g magnetni naboj

induktivnost

a) $\Phi(t) = ?$ $v = 0$

b) $\Phi(t) = ?$

c) $I(t) = ?$

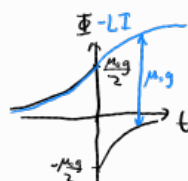
$$\vec{B}_{\text{monopol}} = \frac{\mu_0}{4\pi} \frac{g}{r^2} \frac{\vec{r}}{r}$$

a) $B = \frac{\mu_0}{4\pi} \frac{g}{r^2 + d^2}$ $ds = 2\pi r dr$
 $\Phi = \int \vec{B} \cdot d\vec{S} = \int B_{\perp} dS = \dots = \frac{\mu_0}{2} g \left[1 - \frac{d}{\sqrt{d^2 + L^2}} \right]$

b) leva: $t < 0$ $d = -vt$

desna: $t > 0$ $d = vt$

$$\Phi(t) = \frac{\mu_0 g}{2} \begin{cases} \left[1 + \frac{vt}{\sqrt{d^2 + L^2}} \right]; & t < 0 \\ \left[1 - \frac{vt}{\sqrt{d^2 + L^2}} \right]; & t > 0 \end{cases}$$



c) $U_i = -\frac{d\Phi}{dt} \mu_0 I_m$ magn. tok

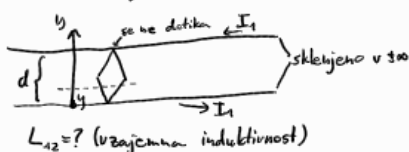
tokovni krog: $U_j = RI + LI + \frac{I}{C}$

$$U_j = LI \quad I_m = g\delta(t)$$

$$LI(t) = -\Phi \Big|_{-\infty}^t - \mu_0 g \theta(t) \Big|_{-\infty}^t$$

$$\boxed{LI(t) = I(t) + \mu_0 g \theta(t)}$$

3. Mag. indukcija v kvadratnem okvirju



$L_{12} = ?$ (vzajemna induktivnost)

$$\Phi_2 = L_{21} I_1 \text{ (definicija)}$$

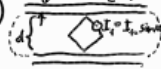
$$L_{ij} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\vec{l}_i \cdot d\vec{l}_j}{|\vec{r}_i - \vec{r}_j|} \Rightarrow L_{ij} = L_{ji}$$

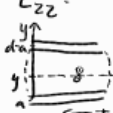
$$B(y) = \frac{\mu_0 I}{2\pi y} + \frac{\mu_0 I}{2\pi(d-y)}$$

$$\Phi = \int B \cdot dS = \frac{\mu_0 I}{2\pi} \int_0^d \left(\frac{1}{y} + \frac{1}{d-y} \right) 2y dy = \frac{2\mu_0 I}{\pi} \int_0^d \frac{d}{d-y} dy = -\frac{2\mu_0 I}{\pi} \ln \frac{d-y}{d} = \boxed{\frac{2\mu_0 I}{\pi} d \ln 2}$$

$$dS = h dy = 2y dy$$

$$\Rightarrow L_{12} = L_{21} = \frac{\Phi_m}{I} = \boxed{\frac{2 \ln 2}{\pi} \mu_0 d}$$

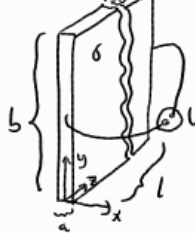
b)  $\Phi_2 = L_{21} I_1$
 $U_{12} = -\dot{\Phi}_2 = -L_{21} \dot{I}_1$
 $U_2 = L_{22} \dot{I}_2$
 $\dot{I}_2 = -\frac{L_{12}}{L_{22}} \dot{I}_1 \rightarrow I_2 = -\frac{L_{12}}{L_{22}} I_1 \sin \omega t \neq$
 I_{20}

$L_{22}:$  $B(y) = \frac{\mu_0}{2\pi} \frac{I}{y} + \frac{\mu_0}{2\pi} \frac{I}{d-y}$
 $\Phi = \int B dS = \int_a^{d-a} \frac{\mu_0}{2\pi} I \left(\frac{1}{y} + \frac{1}{d-y} \right) dy = \dots = \frac{\mu_0}{\pi} I l \ln \frac{d-a}{a} = \frac{\mu_0}{\pi} I l \ln \frac{d}{a}$

$$\rightarrow L_{22} = \frac{\mu_0}{\pi} l \ln \frac{d}{a}$$

$$\rightarrow \frac{I_{20}}{I_{10}} = \frac{L_{21}}{L_{22}} = \frac{2\mu_0 I_1 l \ln \frac{d}{a}}{\frac{\mu_0}{\pi} l \ln \frac{d}{a}} = \frac{2d}{l} \ln \frac{d}{a}$$

4) Kožni pojav u prevodnim trakov



$$a \ll b \ll l$$

$$a) E_z(\omega) = ?$$

$$b) Z(\omega) = \frac{U_0}{I_0} = ?$$

$$j_z = \sigma E_z = \sigma E_0 \frac{\cosh \gamma x}{\cosh \gamma \frac{a}{2}}$$

$$I_0 = \int j_z b dx = \frac{\sigma E_0 b}{\gamma} \frac{\sinh \gamma x}{\cosh \gamma \frac{a}{2}} \Big|_{-\frac{a}{2}}^{\frac{a}{2}} = \frac{\sigma E_0 b}{\gamma} 2 \tanh \gamma \frac{a}{2}$$

$$U = \int_0^l E dz = E_0 l$$

$$\rightarrow Z(i\omega) = \frac{U_0}{I_0} = \frac{E_0 l \gamma}{\sigma b} \coth \gamma \frac{a}{2} = \frac{R_0 \gamma}{2 \sigma b} \coth \gamma \frac{a}{2} = R_0 \frac{\gamma a}{2 \tanh \gamma \frac{a}{2}}$$

$$1) \text{ mali } \omega \Rightarrow \gamma \ll \frac{1}{a} \Rightarrow Z = R_0$$

$$2) \text{ veliki } \omega \Rightarrow \gamma \gg \frac{1}{a}, \gamma a \gg 1 \Rightarrow Z(\omega) = R_0 \frac{\gamma a}{2} = \frac{R_0 a}{2} \frac{i+1}{\sqrt{2}} \sqrt{\mu \sigma \omega}$$

$$\rightarrow R = R_c(z) = \frac{R_0 a}{2\sqrt{2}} \sqrt{\mu \sigma \omega} \propto \sqrt{\omega}$$

Manjka 2 uri

$$\Rightarrow \nabla^2 \vec{E} - \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} = 0$$

$$\vec{E} \propto e^{i\omega t} \Rightarrow \frac{\partial \vec{E}}{\partial t} = i\omega \vec{E}$$

$$\nabla^2 \vec{E} - i\mu_0 \sigma \omega \vec{E} = 0$$

$$\gamma = \sqrt{i\mu_0 \sigma \omega} = \frac{1+i}{\sqrt{2}} \sqrt{\mu_0 \sigma \omega}$$

$$\vec{E} = E_z(x)$$

$$\rightarrow \frac{\partial^2}{\partial x^2} E_z - \gamma^2 E_z = 0$$

$$\text{RP } E_z(\pm \frac{a}{2}) = E_0 \rightarrow \text{sada}$$

$$E_z = A \sinh \gamma x + B \cosh \gamma x$$

$$E_0 = B \cosh \gamma \frac{a}{2} \rightarrow B = \frac{E_0}{\cosh \gamma \frac{a}{2}}$$

$$E_z = E_0 \frac{\cosh \gamma x}{\cosh \gamma \frac{a}{2}}$$

Amplituda